

In this paper, without the explain, $\|\cdot\|$ means $\|\cdot\|_2$, the 2-norm.

$$A = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & & & \vdots \\ \vdots & & & \vdots \\ A_{n1} & \cdots & \cdots & A_{nn} \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} B_{11} & B_{12} & \cdots & B_{1n} \\ B_{21} & & & \vdots \\ \vdots & & & \vdots \\ B_{n1} & \cdots & \cdots & B_{nn} \end{pmatrix}$$

Denote $e_i \in F^{n \times 1}$, then

$$Ae_i = \begin{pmatrix} A_{1i} \\ A_{2i} \\ \vdots \\ A_{ni} \end{pmatrix} \quad \text{and} \quad e_i^T A = (A_{i1} \ A_{i2} \ \cdots \ A_{in})$$

$$AB = \left(\sum_{k=1}^n A_{ik} \cdot B_{kj} \right)_{i,j} = \left(f(A^T e_i, A e_j) \right)_{i,j},$$

where $f: F^n \times F^n \rightarrow F; \left(\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \right) \mapsto \sum_{k=1}^n x_k y_k$, which is the bilinear form on F^n .

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}_{n \times 1} \cdot \begin{bmatrix} b_1 & b_2 & \cdots & b_n \end{bmatrix}_{1 \times n} = \begin{bmatrix} a_1 b_1 & a_1 b_2 & \cdots & a_1 b_n \\ a_2 b_1 & & & \vdots \\ \vdots & & a_{ij} b_j & \vdots \\ a_n b_1 & \cdots & \cdots & a_n b_n \end{bmatrix}$$

Basic Matrix

Def. Let $A \in M_{m \times n}(F)$ be given where $F = \mathbb{R}$ or $\mathbb{C}$. Define the supremum norm of A

$$\|A\|_p \stackrel{\text{def}}{=} \sup_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p} \stackrel{(*)}{=} \sup_{\|x\|_p=1} \|Ax\|_p$$

where $\|\cdot\|_p$ is the p -norm, that is,

$$\|\cdot\|_p: F^n \rightarrow \mathbb{R}; (\alpha_i)_{i=1}^n \mapsto \left(\sum_{i=1}^n |\alpha_i|^p \right)^{\frac{1}{p}}.$$

The equality $(*)$ is given by the fact that for any non-zero vector x , then

$$\frac{\|Ax\|_p}{\|x\|_p} = \left\| A \cdot \frac{x}{\|x\|_p} \right\|_p = \|Ax'\|_p, \text{ where } \|x'\|_p = 1.$$

Prop. Given $A \in M_{m \times n}(F)$, the following conditions are satisfied:

$$1) \|A\|_1 = \max_{1 \leq j \leq n} \|Ae_j\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |A_{ij}|$$

$$2) \|A\|_\infty = \max_{1 \leq i \leq m} \|e_i^T A\|_1 = \max_{1 \leq i \leq m} \sum_{j=1}^n |A_{ij}|$$

$$3) \|A\|_2 = \sigma_{\max}(A), \text{ the largest singular value for } A.$$

Proof. By definition, $\|A\|_p = \sup_{\|x\|_p=1} \|Ax\|_p$ and given $x \in F^n$ s.t. $\|x\|_1 = 1$, ($x = (x_1, \dots, x_n)$)

$$\|Ax\|_1 = \left\| \sum_{k=1}^n x_k A e_k \right\|_1 \leq \sum_{k=1}^n |x_k| \cdot \|A e_k\|_1 \leq \sum_{k=1}^n |x_k| \cdot \max_{1 \leq j \leq n} \|A e_j\|_1 = \max_{1 \leq j \leq n} \|A e_j\|_1$$

And, take $x = e_j$ where the index $1 \leq j \leq n$ satisfying $\|A e_j\|_1$ is the maximal value, we obtain the equality. 1) And, to show the second assertion, observe that

$$\|Ax\|_\infty = \left\| \sum_{k=1}^n x_k A e_k \right\|_\infty \leq \sum_{k=1}^n |x_k| \cdot \|A e_k\|_\infty$$

Differentiation.

Def. Let $A \subseteq \mathbb{R}^n$; let $f: A \rightarrow \mathbb{R}^m$. For $a \in \text{int } A$ and $u \in \mathbb{R}^n$ with $u \neq 0$, the limit

$$f'(a; u) \stackrel{\text{def}}{=} \lim_{t \rightarrow 0} \frac{f(a+tu) - f(a)}{t} \quad (\text{if the limit exists})$$

is called the directional derivative of f at a with respect to u .

And, f is said to be differentiable at a if there is a linear map $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ s.t.

$$\frac{f(a+h) - f(a) - T(h)}{\|h\|} \rightarrow 0 \quad \text{as } h \rightarrow 0. \quad (h \in \mathbb{R}^n)$$

Such a linear map T is unique, and it is denoted $Df_a = T$, called derivative.

Thm. Let $A \subseteq \mathbb{R}^n$; let $f: A \rightarrow \mathbb{R}^m$.

If f is differentiable at a , then all the directional derivatives of f at a exist

$$\text{satisfying } f'(a; u) = Df(a) \cdot u \quad \text{for all } u \in \mathbb{R}^n.$$

Proof By definition, given $u \in \mathbb{R}^n$ with $u \neq 0$,

$$\lim_{t \rightarrow 0} \frac{f(a+tu) - f(a) - Df(a) \cdot (tu)}{\|t \cdot u\|} = 0.$$

Meanwhile, since $Df(a)$ is linear, $Df(a) \cdot (tu) = t \cdot Df(a) \cdot u$.

$$\begin{aligned} \lim_{t \rightarrow 0} \left\| \frac{f(a+tu) - f(a)}{|t| \cdot \|u\|} - \frac{t \cdot Df(a) \cdot u}{|t| \cdot \|u\|} \right\| \\ = \lim_{t \rightarrow 0} \left\| \frac{f(a+tu) - f(a)}{t} - Df(a) \cdot u \right\| \cdot \left| \frac{t}{|t|} \right| \cdot \|u\| = 0 \end{aligned}$$

So proved. $\square$

But, the converse is not true: observe

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}; (x, y) \mapsto \frac{x^2 y}{x^4 + y^2} \quad \text{if } (x, y) \neq (0, 0), \text{ but } (0, 0).$$

Def. Given $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, define the j th partial derivative at a as

$$D_j f(a) \stackrel{\text{def}}{=} f'(a, e_j).$$

Thm. Given $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, denote $f_i = \pi_i \circ f: \mathbb{R}^n \rightarrow \mathbb{R}$. ($1 \leq i \leq m$)
 If f_i is differentiable at a , then

$$Df_i(a) = [D_1 f_i(a) \quad D_2 f_i(a) \quad \dots \quad D_n f_i(a)]$$

Moreover,

f is differentiable at a if and only if for each f_i is differentiable at a .

Proof. The first assertion is directly, by

$$Df_i(a) \cdot e_j \stackrel{\text{Lemma}}{=} f_i(a; e_j) \stackrel{\text{Def.}}{=} D_j f_i(a)$$

And, observe that: let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be any linear map. Consider a map

$$F(h) = \frac{f(a+h) - f(a) - T \cdot h}{\|h\|} \quad \text{for small enough } h \in \mathbb{R}^n.$$

Since the projection is linear,

$$\pi_i \circ F(h) = \frac{f_i(a+h) - f_i(a) - \pi_i \circ T \cdot h}{\|h\|}$$

By the topological property, the limit $\lim_{h \rightarrow 0} F(h)$ exists iff $\pi_i \circ F(h) \rightarrow 0$ for each i .

Thus the first assertion gives that the result. Visually,

$$Df(a) = \left(D_j f_i(a) \right)_{i,j} = \begin{bmatrix} D_1 f_1(a) & D_2 f_1(a) & \dots & D_n f_1(a) \\ D_1 f_2(a) & & & \vdots \\ \vdots & & \ddots & \vdots \\ D_1 f_m(a) & \dots & \dots & D_n f_m(a) \end{bmatrix}$$

$$\left(= \frac{\partial f_i}{\partial x_j}(a) \right)$$

Thm. Let A be an open subset in $\mathbb{R}^n$, and let $f: A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^3$.
 Suppose that for all $1 \leq i \leq m$, $1 \leq j \leq n$, the partial derivatives $D_j f_i$ exists and continuous.
 Then, f is differentiable. (Such a function is called of class C^1).
 Proof. Since f is differentiable iff for each i , f_i is differentiable,
 so given $1 \leq i \leq m$, showing f_i is differentiable under the condition is enough.
 Let $a \in A$ be given. Given $h = (h_1, \dots, h_n) \in \mathbb{R}^n$, Consider

$$p_0 = a$$

$$p_1 = a + h_1 \cdot e_1$$

$$p_2 = a + h_1 \cdot e_1 + h_2 \cdot e_2$$

$\vdots$

$$p_n = a + h_1 e_1 + \dots + h_n e_n = a + h.$$

Then, by the telescoping,

$$f_i(a+h) - f_i(a) = \sum_{j=1}^n [f_i(p_j) - f_i(p_{j-1})]$$

And, for fixed $1 \leq j \leq n$,

$$f_i(p_j) - f_i(p_{j-1}) = f_i(p_{j-1} + h_j e_j) - f_i(p_{j-1}) = D_j f_i(p_{j-1} + g_j e_j) \cdot h_j$$

where for some g_j , between 0 and h_j , by the Mean-Value Theorem.

(If $h_j = 0$, still the equality holds.). So,

$$f_i(a+h) - f_i(a) = \sum_{j=1}^n D_j f_i(p_{j-1} + g_j e_j) \cdot h_j.$$

And, since for $B = [D_1 f_i(a) \dots D_n f_i(a)]$,

$$\frac{f_i(a+h) - f_i(a) - Bh}{\|h\|} = \frac{\sum_{j=1}^n [D_j f_i(p_{j-1} + g_j e_j) - D_j f_i(a)] \cdot h_j}{\|h\|} \rightarrow 0 \text{ as } h \rightarrow 0.$$

Chain Rule.

Thm. Let $A \subseteq \mathbb{R}^n$ and $B \subseteq \mathbb{R}^m$ be open and $f: A \rightarrow \mathbb{R}^m$ and $g: B \rightarrow \mathbb{R}^p$ with $f[A] \subseteq B$. If f is differentiable at $a \in A$ and g is differentiable at $f(a) \in B$, then the composite $g \circ f$ is differentiable at a with

$$D(g \circ f)(a) = Dg(b) \cdot Df(a).$$

Proof. Since B is open, choose $\epsilon > 0$ such that $B_\epsilon(f(a)) \subseteq B$.

And, since f is continuous at a , there is $\delta > 0$ such that

$$f[B_\delta(a)] \subseteq B_\epsilon(f(a))$$

In other words, given small enough $\epsilon > 0$, there is $\delta > 0$ such that

$$\|x - a\| < \delta \Rightarrow \|f(x) - f(a)\| < \epsilon.$$

So that, the composition $g \circ f$ is well-defined on $B_\delta(a) \subseteq A$.

Let $\Delta(h) = f(a+h) - f(a)$ for $\|h\| < \delta$. And, define

$$F(h) = \frac{\Delta(h) - Df(a) \cdot h}{\|h\|} \quad \text{for } 0 < \|h\| < \delta.$$

Since f is differentiable at a , by definition, $F(h) \rightarrow 0$ as $h \rightarrow 0$. So, since observing that

$$\|\Delta(h)\| = \|Df(a) \cdot h + \|h\| \cdot F(h)\| \leq \|Df(a)\| \cdot \|h\| + \|F(h)\| \cdot \|h\| = [\|Df(a)\| + \|F(h)\|] \cdot \|h\|.$$

Since $F(h)$ converges, the quotient $\|\Delta(h)\|/\|h\|$ is bounded in $0 < \|h\| < \delta$.

Now, define G as $G(0) = 0$ and

$$G(k) = \frac{g(f(a) + k) - g(f(a)) - Dg(b) \cdot k}{\|k\|} \quad \text{for } 0 < \|k\| < \epsilon.$$

So that for all $\|k\| < \epsilon$,

$$g(f(a) + k) - g(f(a)) = Dg(b) \cdot k + \|k\| \cdot G(k).$$

Consequently, substituting $k = \Delta(h)$ and by the fact that $f(a+h) = f(a) + \Delta(h)$,

$$\frac{1}{\|h\|} \cdot [g(f(a+h)) - g(f(a)) - Dg(b) \cdot Df(a) \cdot h] = \frac{1}{\|h\|} \cdot [Dg(b) \cdot h$$